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TAKE HOME PROJECT

DEADLINE: NOVEMBER 7TH, NOON

OBJECTIVE

To create a solution capable of extracting structured information from PDF files synchronously, with the best possible cost-benefit ratio and accuracy.

GENERAL DESCRIPTION

You will need to build a solution that will be used to **extract structured data from thousands of documents,** and you must:

• Respond to each request in less than 10 seconds; • Minimize the monetary cost of execution as much as possible; • Guarantee consistency and accuracy of the extracted entities (all responses are (unique for each document).

For each document, there are three input parameters: ÿ identifies

1. label the document type (e.g., "invoice", "contract", "form").
2. extraction_schema ÿ a dictionary where the keys are the names of the fields to be extracted and the values are descriptions of the fields. (i.e., “cpf”: “individual taxpayer identification number in the format XXX.XXX.XXX-X”)
3. pdf ÿ the PDF file

The goal is to extract the information from the PDF in the format defined by the schema. Your system can accumulate knowledge with each request.

Problem limitations: • Each PDF has only one page. • The PDF already has OCR done, meaning all the text is embedded in the page. • You don't know the labels or their respective schemas beforehand. complete.

• For each label, its corresponding complete schema is fixed. • The fields of the schema specified as a document parameter belong to the label's complete schema. extraction_schema

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Example: consider a fictitious label "RG" with a complete schema. {name, mother's_name, father's_name, ID_number, birth_date} Remember: you don't He knows the complete schematic.

- In your first request, you may receive a document with the label "RG" and extraction_schema = {name, mother's_name, birth_date}
- In a second request, you can receive a document. with label "RG" and extraction_schema = {name, father's_name, ID_number, birth_date}

- Documents with the same label follow a standard template, but:
  - Some (like forms) have an identical structure across files;
  - Others (such as contracts or invoices) maintain the general layout, but the Information may vary depending on location.

DATA FOR DEVELOPMENT

This repository It has some examples of documents with their labels and extraction_schemas corresponding labels. You will find examples of three labels. different, and you can use them freely to train, test, and adjust your solution.

DELIVERABLE

- All code on Github or equivalent.
- A README.md file explaining:
  - What challenges did you identify, which ones did you decide to address, and what is your solution? proposal for each of them - here is the space for you to put in Words represent all the creativity you applied.
  - How to use your solution.
- Have a solution that receives a request with (label, schema, pdf) and returns the Structured extracted data.

You can deliver your solution in any functional format, such as:

- An API endpoint (e.g., /extract)
- A simple web interface (UI)
- An executable script

WHAT INTERESTS US

Extracting data with LLMs has become a commodity. We want to see **technical creativity** to...

Develop effective strategies that:

1. Reduce calls to LLM (cache, pattern matching, local OCR parsing)
2. Maintain 80%+ accuracy even with variable layouts.

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3. Respond in <10 seconds per request. 4. Minimize monetary cost.

Examples of interesting challenges you might want to address: • Balancing accuracy vs. cost: the trade-off between using fast heuristics and LLM.

Examples: •

Creative solutions to reduce costs (e.g., preprocessing, caching, heuristics, etc.) • How to use LLMs and other NLP techniques

• Optimize the context.

• Dealing with variability: documents with the same label may have different layouts.

EVALUATION CRITERIA

We have developed an internal dataset containing thousands of examples that will be used to evaluate your solution. For each item, we will assess your solution and measure:

• Response time: extractions/responses should be returned, on average, in less than 10 seconds.

• Accuracy: At least 80% of the fields must be extracted correctly:

• 1 incorrect character = incorrect field • Evaluation is not case sensitive (ENTER = enter = EnTer != inter) • Cost: What was the average cost to extract each document?

Key differentiators

In addition to the factors mentioned above, we will also take into consideration any attempt to go beyond the obvious scope foreseen, such as:

• A functional and intuitive UI

• Exceptional speed (extractions in less than 0.1 seconds)

Infrastructure and Restrictions

• You will receive an OpenAI API key with a budget limit. • The credits are sufficient for the challenge, but you can request more if needed.

more.

• For LLM calls, use only the gpt-5 mini model (link)

Tip: the cost of calls to the LLM is the upper bound of your system's total cost — optimizing model usage is an essential part of the challenge.

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FAQ

1. Do I need to train a model?

No. The focus is on building the end-to-end solution creatively, not on...
model training.

2. I need to worry about extracting the Text/perform OCR on documents?

No.

3. I can use any languages. or External libraries?

Yes - any language, library, or framework is allowed.

4. Can I use embeddings? or other NLP techniques?

Yes, as long as the execution remains within the cost limit.

5. The `extraction_schema` request exists if there is a field that... Not a PDF?

Return null for this field.

6. Can I cache PDF results? Already processed?

Yes, as long as it's only between requests within the same session.

7. If I call me one LLM, will count in time 10s?

The time to be considered will be the time elapsed since the action considered as the
From sending to receiving the response. In other words, what happens asynchronously and in
Background is irrelevant.

8. The labels on evaluations will be the same as the of development?

Not necessarily. Your system should be adaptive. Remember: you don't know
neither the labels nor the extraction_schema in advance.

9. How should I submit it? the solution?

ÿ GitHub repository with a README detailing its approach and implementation instructions.

10. Where are the data for I Test my solution?

ÿ They are in the following GitHub repository.

10. Do I need to create an API?

Any format is acceptable, as long as it is easy to use. For example:

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1. Native application: Must allow batch parameter passing, specifying the path to the folder containing the files.
2. Web application + API: Must allow batch parameter passing. specifying the path to the folder containing the files.
3. Script / CLI: Must allow batch parameter passing, specifying the path to the folder containing the files.

It is important that batch processing be done serially, meaning each line should be processed independently of the following ones. The first input of a batch should be returned in less than 10 seconds.

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EXAMPLE 1

Label: OAB_wallet

-> PDF:

-> Schema:

```
JSON
{
    "Name": "Name of the professional, usually in the upper left corner of the image", "Registration": "Professional's registration
    number", "Section": "Section of the professional", "Subsection": "Subsection to
    which the professional belongs", "Category": "Category, can be
    LAWYER, SUPPLEMENTARY,

INTERN, FEMALE INTERN
    "telefone_profissional": "Professional's phone number", "situacao": "Professional's
    status, usually in the bottom right corner."

}
```

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Output:

JSON


```
{
    "name": "SON GOKU",
    "registration": "101943",
    "section": "PR",
    "subsection": "Sectional Council - Paraná", "category":
    "Supplementary", "professional_phone":
    null, "status": "Regular Status"

}
```

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EXAMPLE 2

Label: OAB_wallet

-> Schema:

```
JSON
{
    "Name": "Name of the professional, usually in the upper left corner of the image", "Registration": "Professional's registration
    number", "Section": "Section of the professional", "Subsection": "Subsection to
    which the professional belongs", "Category": "Category, can be
    LAWYER, SUPPLEMENTARY,

INTERN, FEMALE INTERN
    "endereco_profissional": "Full professional address", "situacao": "Professional status, usually in
    the lower right corner."

}
```

-> PDF:

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Output:

JSON


```
{
    "name": "JOANA D'ARC",
    "registration": "101943",
    "section": "PR",
    "subsection": "Sectional Council - Paraná", "category":
    "Supplementary",
    "professional_address": "Avenida Paulista, Nº 2300, ground floor, Pilotis, Bela
Vista, São Paulo - SP, 01310300", "situacao":
    "Regular Status"
}
```